

MODEL QUESTION SOLUTION

It mainly contains numerical part. Theory part were already cover in the book.

2023 Spring

1. (b) Graphical systems are composed of DAC and Frame Buffer organization. Why? Explain the working principle of Data gloves with example

Solution:

DAC (Digital-to-Analog Converter):

- **Analog Signal Conversion:** Converts digital pixel values into analog voltage levels.
- **Compatibility:** Essential for compatibility with analog display devices like CRT monitors and some audio equipment.
- **Color Precision:** Provides precise control over color and intensity for each pixel.
- **Image Quality:** Ensures smooth transitions and accurate color representation on analog displays.
- **Legacy Use:** While digital displays are prevalent today, DACs are still relevant for legacy equipment and specialized applications.
- **Signal Clarity:** Reduces signal noise and distortion, resulting in a cleaner image on analog displays.
- **Realism:** Contributes to the realism of graphics by enabling smooth gradients and color blending.

Frame Buffer:

- **Pixel Data Storage:** Dedicated memory area to store pixel data for the current frame being displayed.
- **Double Buffering:** Often used in double buffering techniques to prevent visual artifacts like flickering and tearing in animations.
- **Efficient Updates:** Provides a stable mechanism for making changes to the display by updating the frame buffer, improving efficiency.
- **Resolution and Color Depth:** Frame buffer size corresponds to the display's resolution and color depth (e.g., 1080p, 32-bit color).
- **Performance Optimization:** Allows the CPU and GPU to work asynchronously, with the CPU preparing the next frame in the frame buffer while the current frame is displayed.

- **Z-Buffer:** Can include a depth (z-buffer) component for 3D graphics, helping determine which objects are visible and ensuring correct rendering order.
 - **Critical for Graphics Rendering:** The frame buffer is a fundamental component in graphics rendering, ensuring the smooth and efficient display of images and animations.
2. (b) Why circle and ellipse uses different point symmetry? Explain the mid-point circle generation algorithm by calculating its initial decision parameter.

Solution:

Because they are defined by various geometrical equations and geometrical features, circles and ellipses employ different point symmetries. The relationship between a shape's equation and its center determines point symmetry, often referred to as rotational symmetry.

Circle:

- The collection of all points in a plane that are equally spaced from a central point known as the center is referred to as a circle.
- The formula for a circle with a radius of 'r' and a center at (0,0) is provided by: $x^2 + y^2 = r^2$.
- The fact that a circle is symmetrical with regard to its center is its primary characteristic. This indicates that a circle remains intact no matter what angle is rotated around its center. It's called rotational symmetry.

Ellipse

- An ellipse is described as the collection of all points on a plane where the total of the distances between any two fixed points (referred to as foci) remains constant.
- The equation for an ellipse with its major axis parallel to the x-axis and its center at (0, 0) is $(x^2/a^2) + (y^2/b^2) = 1$, where 'a' and 'b' are the semi-major and semi-minor axes, respectively.
- The fact that an ellipse possesses two axes of symmetry—a major axis and a minor axis—is its primary characteristic. Due to its symmetry with regard to both axes, it does not change whether you reflect or spin it at specific angles from its center. This leads to two lines of symmetry (vertical and horizontal), which are different from rotational symmetry.

In conclusion, due of the differences in their definitions and equations, circles and ellipses have different point symmetries.

While ellipses feature both horizontal and vertical lines of symmetry because of their two axes of symmetry, circles display rotational symmetry with regard to their center. These variations result from the mathematical definitions of these forms.

5. (a) Define blobby objects with example. What is cubic Bezier curve? Derive the equation of Cubic Bezier curve.

Solution:

- Blobby objects are a type of geometric primitive in computer graphics.
- They are used to represent amorphous or organic shapes, like clouds, smoke, and fluids.
- Blobby objects are associated with the concept of "metaballs."
- Metaballs are mathematical functions defining a field of influence around points in space.
- When metaballs' influence fields overlap, they blend together to create a smooth, blob-like surface.
- The resulting surface lacks a fixed polygonal structure and can deform and merge smoothly with other blobby objects.
- Metaballs are commonly used in CGI for modeling liquids, soft body physics simulations, and special effects.



2022 Fall

1. (b) Define Refresh rate and Resolution. Consider a raster scan system having 20-inch by 30-inch screen with a resolution of 300 pixels per inch in each direction. If the display controller of the system refreshes the screen at a rate of 60 frames per second, how many pixels could be accessed per second and what is the access time per pixels of the system.

Solution:

$$\text{Total pixel} = 20 \times 300 \times 30 \times 300$$

$$\text{Refresh rate} = 60 \text{ FPS.}$$

$$\text{Pixel accessed per sec (F)} = 20 \times 300 \times 30 \times 300 \times 60 \\ = 3240000000$$

$$\text{Access time per pixel} = \frac{1}{F}$$

$$= \frac{1}{3240000000} \\ = 308.64 \times 10^{-12} \text{ sec.}$$

2. (b) Consider a line from (2,1) to (8,3) using DDA algorithm to rasterize a line.

Solution:

$$(x_1, y_1) = (2, 1)$$

$$(x_2, y_2) = (8, 3)$$

$$\Delta x = x_2 - x_1 = 8 - 2 = 6$$

$$\therefore \Delta x > \Delta y$$

$$\Delta y = y_2 - y_1 = 3 - 1 = 2$$

$$\therefore 2\Delta y = y, 2\Delta x = 12$$

$$m = \frac{\Delta y}{\Delta x} = \frac{2}{6} = \frac{1}{3} < 1$$

Hint:

$$d_1 - d_2 > 0$$

$$y_{k+1} = y_k + 1$$

$$x_{k+1} = x_k + 1$$

$$P_{k+1} = P_x + 2\Delta y - 2\Delta x$$

$$d_1 - d_2 < 0$$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$P_{k+1} = P_k + 2\Delta y$$

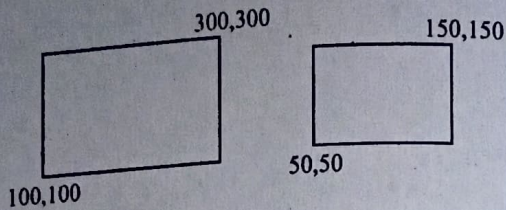
Initial decision parameter,

$$P_0 = 2\Delta y - \Delta x = 4 - 6 = -2$$

n	P_k	x_{k+1}	y_{k+1}
-	-	2	1
0	-2	3	1
1	2	4	2
2	-6	5	2
3	-2	6	3
4	2	7	3
5	-6	8	3

4. (a) Window port is given by (100, 100, 300, 300) and viewport is given by (50, 50, 150, 150). Convert the window port coordinate (200, 200) to viewport coordinate.

Solution:



Here,

$$X_{wmin} = 100$$

$$Y_{wmin} = 50$$

$$X_{wmax} = 300$$

$$Y_{wmax} = 300$$

$$X_{vmin} = 100$$

$$Y_{vmin} = 50$$

$$X_{vmax} = 300$$

$$Y_{vmax} = 300$$

$$S_x = \frac{\text{width of viewport}}{\text{width of window}}$$

$$= \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}}$$

$$= \frac{150 - 50}{300 - 100}$$

$$= \frac{100}{200} = \frac{1}{2}$$

$$S_y = \frac{\text{height of viewport}}{\text{height of window}}$$

$$= \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}}$$

$$= \frac{150 - 50}{300 - 100} = \frac{1}{2}$$

Window co-ordinate

$$(X_w, Y_w) = 200, 200$$

$$\therefore X_v = X_{vmin} + (X_w - X_{wmin}) \times S_x$$

$$= 50 + (200 - 100) \times \frac{1}{2}$$

$$= 50 + 50$$

$$= 100$$

$$Y_v = Y_{vmin} + (Y_w - Y_{wmin}) \times S_y$$

$$= 50 + (200 - 100) \times \frac{1}{2}$$

$$= 50 + 50$$

$$= 100$$

$\therefore$ View port co-ordinates are (100, 100).

- (b) A mirror is placed vertically such that it passes through the point (5, 0) and (0, 5). Find the reflected view of triangle ABC with coordinates A(5, 30), B(30, 50) and C(20, 60).

Solution:

$$x_1 = 5$$

$$x_2 = 0$$

$$y_1 = 0$$

$$y_2 = 5$$

$$A(5, 30), B(30, 50), C(20, 60)$$

$$\text{Slope (m)} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 5}{5 - 0} = -1$$

$$\text{Rotation angle } (\theta) = -1$$

$$\therefore \theta = \tan^{-1}(-1) = -45^\circ$$

Steps:

1. Translation by $T_x = -5$ or $T_x = 0$

$$T_y = 0$$

$$T_y = -5$$

both is possible,

$$T_{[-5, 0]} = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Counter clockwise rotation with $\theta = 45^\circ$

$$R_{\theta} = 45^\circ \text{ CCW} = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Reflection x-axis

$$\text{Ref}(x\text{-axis}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Reverse rotation $R_{\theta}^{-1} = 45^\circ$

$$R_0^{-1} = 45^\circ \text{ CW} = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Anti translation by $T_x = 5, T_y = 0$

$$T_{(5,0)}^{-1} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$CM = T_{(5,0)}^{-1} R_0^{-1} = 45^\circ \text{ CW Ref}(x\text{-axis}) R_0 = 45^\circ \text{ CCW } T_{(-5,0)}$$

= Calculate yourself.

$$\therefore P' = CM \times P = CM \times \begin{bmatrix} 5 & 30 & 20 \\ 30 & 50 & 60 \\ 1 & 1 & 1 \end{bmatrix}$$

6. (a) Derive the equation for cubic Bezier curve and find the coordinate at $t=0.2$ with respect to the control points (1,1), (4,6), (8,-3) and (12,2).

Solution:

Control point = y

Degree of polynomial = $n - 1 = 3$

We know the Beizer cubic polynomial equation.

$$P(t) = (1-t)^3 P_0 + 3t(1-t)^2 P_1 + 3t^2(1-t) P_2 + t^3 P_3 \quad 0 \leq t \leq 1$$

Control points

$$A(1,1) = P_0(x_0, y_0)$$

$$B(4,6) = P_1(x_1, y_1)$$

$$C(8,-3) = P_2(x_2, y_2)$$

$$D(12,2) = P_3(x_3, y_3)$$

Let $t = 0.2$

$$\begin{aligned} x(0.2) &= x_0(1-t)^3 + 3t(1-t)^2 x_1 + 3t^2(1-t) x_2 + t^3 x_3 \\ &= 1(1-0.2)^3 + 3 \times 0.2(1-0.2)^2 \times 4 + 3 \times 0.2^2(1-0.2) \times 8 + 0.2^3 \times 12 \\ &= 2.912 \end{aligned}$$

$$\begin{aligned} y(0.2) &= (1-t)^3 y_0 + 3t(1-t)^2 y_1 + 3t^2(1-t) y_2 + t^3 y_3 \\ &= (1-0.2)^3 \times 1 + 3 \times 0.2(1-0.2)^2 \times 6 + 3 \times 0.2^2(1-0.2) \times (-3) + 0.2^3 \times 2 \\ &= 2.544 \end{aligned}$$

Alternative

It can be further be simplified and written in matrix form as

$$P(t) = [t^3 \ t^2 \ t^1] \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_0 \\ P_1 \\ P_2 \\ P_3 \end{bmatrix}$$

At $t = 0.2$

$$= [0.2^3 \ 0.2^2 \ 0.2 \ 1] \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 0 \\ 8 & -3 & 0 \\ 12 & 2 & 0 \end{bmatrix}$$

$$= [0.2^3 \ 0.2^2 \ 0.2 \ 1] \begin{bmatrix} -1 & 28 & 0 \\ 3 & -42 & 0 \\ 9 & 15 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= [2.912 \ 2.544 \ 0]$$

2021 Spring

2. (a) Write Bresenham's line drawing algorithm along with necessary derivation for positive slope less than 1 i.e. $|m| < 1$. Trace the algorithm for the line with end points A(6,12) and B(10,15).

Solution:

End points A(6,12), B(10,15)

$$\text{Slope } M = \frac{y_2 - y_1}{x_2 - x_1} = \frac{15 - 12}{10 - 6} = \frac{3}{4} < 1$$

$$x_1 = 6 \quad x_2 = 10$$

$$y_1 = 12 \quad y_2 = 15$$

This is the case of $m < 1$ ($\Delta x > \Delta y$)

tips

$$d_1 - d_2 > 0$$

$$d_1 - d_2 < 0$$

$$y_{k+1} = y_k + 1$$

$$x_{k+1} = x_k + 1$$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x$$

$$P_{k+1} = P_k + 2\Delta y$$

Initial decision parameter

$$P_0 = 2\Delta y - \Delta x$$

$$\Delta x = 10 - 6 = 4$$

$$2\Delta x = 8$$

$$\Delta y = 15 - 12 = 3$$

Initial decision parameter:

$$P_0 = 2\Delta y - \Delta x = 6 - 4 = 2$$

$$2\Delta y = 6$$

n	P _k	X _{k+1}	Y _{k+1}
-	-	6	12
0	2	7	13
1	0	8	14
2	-2	9	14
3	-4	10	15

3. (a) Derive a composite transformation matrix for reflecting an object about a line $y = x + 4$ in 2D.

Solution:

$$y = x + 4 \text{ in 2D}$$

x	0	1	2
y	4	5	6

$$\text{Slope } m = \tan\theta = 1$$

$$\theta = 45^\circ$$

Required steps:

1. Translate the line to origin by decreasing the y-intercept with are:

$$T_{(0,-c)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Rotate the line by angle 45° in clockwise direction so that given line must overlap x-axis.

$$R_{\theta=45^\circ\text{CW}} = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Reflect the object about x-axis,

$$\text{Ref}_{(x\text{-axis})} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Reverse rotate the line by angle 45° in counter clockwise direction

$$R_{\theta=45^\circ\text{CCW}}^{-1} = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Reverse translate the line to original position by adding the y-intercept with form

$$T_{(0,c)}^1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{Composite matrix} = T_{(0,c)}^1 R_{\theta=45^\circ\text{CCW}}^{-1} \text{Ref}_{(x)} R_{\theta=45^\circ} T_{(0,c)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -2\sqrt{2} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -2\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -2\sqrt{2} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & -2\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & -4 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & -4 \\ 1 & 0 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

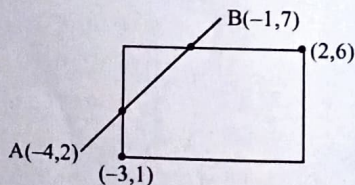
We can directly calculated as

$$\begin{bmatrix} -\frac{(m^2-1)}{m^2+1} & \frac{2m}{m^2+1} & -\frac{2mc}{m^2+1} \\ \frac{2m}{m^2+1} & \frac{m^2-1}{m^2+1} & \frac{2c}{m^2+1} \\ 0 & 0 & 1 \end{bmatrix} m=1, c=4$$

$$= \begin{bmatrix} 0 & 1 & -4 \\ 1 & 0 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

2. (b) Let R be the rectangular window whose lower left hand corner is at L (-3,1) and upper right-hand corner is at (2,6). Use Cohen-Sutherland algorithm to clip the line segments A(-4,2) and B(-1,7).

Solution:



$$A(-4,2) (x_1, y_1)$$

$$B(-1,7) (x_2, y_2)$$

$$\text{Here, } x_{wmin}, y_{wmin} = (-3, 1)$$

$$x_{wmax}, y_{wmax} = (2, 6)$$

$$\text{Slope (m)} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7 - 2}{-1 - (-4)} = \frac{5}{-1 + 4} = \frac{5}{3}$$

Step 1: Region code

For A and B

$$A(-4,2)$$

$$B(-1,7)$$

$$-4 < -3 = 1 \text{ L}$$

$$-4 > 2 = 0 \text{ R}$$

$$2 < 1 = 0 \text{ B}$$

$$2 > 6 = 0 \text{ T}$$

$$\text{Region code of A} = 0001$$

$$\text{Region code of B} = 1000$$

Step 2: Logical anding of A and B.

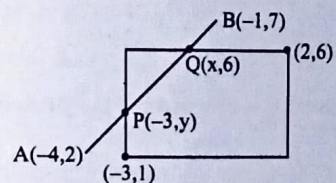
$$A = 0001$$

$$B = 1000$$

$$0000$$

This is the case of partially visible, it might have intersected the window to same point.

Step 3:



We have to find the intersection point of line AB.

For P:

Since P cuts the window at left side so its x_{wmin} remains constant and y changes.

$$x_{wmin} = -3$$

y to find

$$\text{So } A(-4,2) \text{ and } P(-3,y)$$

$$\text{So, slope of AP} = \text{Slope of AB.}$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{5}{3}$$

$$\Rightarrow \frac{y - 2}{-3 + 4} = \frac{5}{3}$$

$$\Rightarrow \frac{y - 2}{1} = \frac{5}{3}$$

$$\Rightarrow y = \frac{5}{3} + 2 = \frac{11}{3} = 3.667$$

$$\therefore P(-3, 3.667)$$

For Q:

Since Q cuts the window at left side so its $y_{\max}$ remains constant and x changes.

$$y_{\max} = 6$$

x to find

So, A(-3,1) and Q(x,6)

Since slope of AQ = Slope of AB so

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{5}{3}$$

$$\frac{6 - 1}{x + 3} = \frac{5}{3}$$

$$\frac{5}{x + 3} = \frac{5}{3}$$

$$3 = x + 3$$

$$\therefore x = 0$$

So, Q(0,6)

Hence the line PQ is visible and the line AP and QB were clipped away or cutoff.

2021 Fall

In a true color system having resolution of 1024×768 having the refresh rate of 60 fps calculate the following:

- Size of frame buffer
- Access time of one frame
- Access time for one pixel
- Access time for one row

Note: Convert your memory into Mega Byte.

Solution:

Digitize the first octant of a circle having radius $r = 8$ and centered at (3,4).

What is 3D transformation? Rotate the triangle A(0,0), B(2,2), C(4,2) about the origin by an angle of 45° .

True color system

Resolution = 1024×768

Refresh rate = 60 fps

- Size of frame buffer = ?
- Access time of one frame = ?

c. Access time for one pixel = ?

d. Access time for one row = ?

a. Since given system is true color system in which color depth is typically 24 bits per pixel (8 bits of each of RGB).

We know,

Size of frame buffer = Resolution $\times$ color depth

$$= 1024 \times 768 \times 24 \text{ bits}$$

$$= 18874368 \text{ bits}$$

$$= 2359296 \text{ byte [1 byte = 8 bits]}$$

$$= 2304 \text{ kb [1 kb = 1024 byte]}$$

$$= 2.25 \text{ mb [1 mb = 1024 kb]}$$

$\therefore$ Size of frame buffer = 2.25 mb

b. Here, refresh rate = 60 fps

60 frames need 1 sec.

$$\therefore 1 \text{ frames need } \frac{1}{60} \text{ sec} = 0.0167 \text{ sec} = 16.7 \text{ ms}$$

$\therefore$ Access time of one frame = 16.7 ms or 0.0167 sec.

c. Since,

$$1 \text{ frame} = 1024 \times 768 \text{ pixel} \\ = 786432 \text{ no. of pixel}$$

$$\text{So, 1 frame needs } \frac{1}{60} \text{ sec.}$$

$$786432 \text{ pixel needs } \frac{1}{60} \text{ sec.}$$

$$\therefore 1 \text{ pixel needs } \frac{1}{60 \times 786432} \text{ sec.}$$

$$= 0.00000002119 \text{ sec.} = 21.19 \text{ ns}$$

$$\therefore \text{Access time for 1 pixel} = 21.19 \text{ ns}$$

d. Here, system contains 768 scan lines and each scan line contains 1024 pixels. [row scan lines]

$$\text{Access time for are row} = \text{Access time of } \frac{\text{One frame}}{\text{Resolution (height)}}$$

$$1 \text{ frame takes } \frac{1}{60} \text{ sec.}$$

$$1 \text{ frame contains 768 scan line}$$

$$768 \text{ scan line takes } = \frac{1}{60} \text{ sec.}$$

$\therefore$ 1 scan line (row) takes $\frac{1}{60 \times 768} \text{ sec} = 21.17 \mu\text{s}$

$\therefore$ Access the for the row 21.17 μs

Here, radius = 8

Center = 3,4

We start from a point (0,6) = (0,8)

$x_0, y_0 = 0,8$

When center is not origin, we first calculate the octants points of the circle in the same way as the center at origin then add center on each calculated pixel.

Initial decision parameter (d_0) = $1 - r = 1 - 8 = -7$

Case (1)

$d_i < 0$

$d_{i+1} = d_i + 2x_i + 3 \quad x_{i+1} = x_i + 1$

or, $y_{i+1} = y_i$

$d_{i+1} = d_i + 2x_{i+1} + 1$

Case (2)

$d_i \geq 0$

$d_{i+1} = d_i + 2(x_i - y_i) + 5 \quad x_{i+1} = x_i + 1$

or, $y_{i+1} = y_i - 1$

$d_{i+1} = d_i + 2x_{i+1} - 2y_{i+1} + 1$

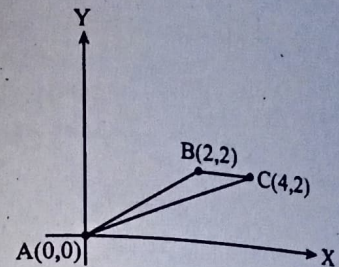
n	d_k	x_{i+1}	y_{i+1}	Plot	(3,4) centre
0	-7	1	8	(1,8)	(4,12)
1	$-7 + 2 \times 1 + 1 = -4$	2	8	(2,8)	(5,12)
2	$-4 + 2 \times 2 + 1 = 1$	3	7	(3,7)	(6,11)
3	$1 + 2 \times 3 - 2 \times 7 + 1 = -6$	4	7	(4,7)	(7,11)
4	$-6 + 2 \times 4 + 1 = 3$	5	6	(5,6)	(8,10)
5	$3 + 2 \times 5 - 2 \times 6 + 1 = 2$	6	5	(6,5)	(9,9)

Discard and stop $x > y$.

Rotate A triangle, A(0,0), B(2,7), C(4,7) by 45° about origin.

Point A(0,0), B(2,1), C(4,2)

Direction is not given so suppose CCW.



We know,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\therefore A' = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$B' = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 2\sqrt{2} \end{bmatrix}$$

$$C' = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ 3\sqrt{2} \end{bmatrix}$$

$\therefore A' = (0,0)$

$B' = (0, 2\sqrt{2})$

$C' = (\sqrt{2}, 3\sqrt{2})$

2020 Fall 2018 Spring

Rasterize the points of given line end points A(-2,-4) and B(-6,-9) using Brenham's line drawing algorithm.

A(-2,-4), B(-6,-1)

Solution:

This is the case of line from right to left

Initial decision parameter

$$P_0 = \Delta y - 2\Delta x$$

$$x_1 = -2 \quad x_2 = -6$$

$$y_1 = -4 \quad y_2 = -9$$

$$\therefore \Delta y = -9 - (-4) = -5$$

$$2\Delta y = -10$$

$$\Delta x = -6 - (-2) = -4$$

$$2\Delta x = -8$$

$$P_0 = \Delta y - 2\Delta x = -5 + 8 = 3$$

Case I

$$P_k < 0$$

$$x_{k+1} = x_k$$

$$y_{k+1} = y_k - 1$$

$$P_{k+1} = P_k - 2\Delta x$$

Case II

$$P_k \geq 0$$

$$x_{k+1} = x_k - 1$$

$$y_{k+1} = y_k - 1$$

$$P_{k+1} = P_k - 2\Delta x + 2\Delta y$$

n	P _i	X _{i+1}	Y _{i+1}
-	-	-2	-4
0	3	-3	-5
1	1	-4	-6
2	-1	-4	-7
3	7	-5	-8
4	5	-6	-9

2019 Fall

Consider a non-interlaced raster monitor with a resolution of 1280×1024. If horizontal and vertical retrace times are 20 microsecond each, then calculate the fraction of the total refresh time per frame spent in retrace of the electron beam? Assume refresh rate of 60 frames per second.

OR,

Explain symmetrical property of circle. Write midpoint circle algorithm and apply that algorithm to find the pixel values of the circle whose radius $r = 10$ and centre of the circle = (0,0).

- Define decision parameter in Bresenham's line drawing? Digitize a circle $(x-2)^2 + (y-3)^2 = 25$ using a midpoint circle drawing algorithm.
- Determine window to viewport transformation matrix for window (5,10), (15,20) and for viewport (8,12) (12,18). Note the coordinates values are for lower left and upper right corner.

Solution:

$$\text{Resolution} = 1280 \times 1024$$

Horizontal and vertical retrace time = 20 μ s.
Total refresh time per frame = ?
Refresh rate = 60 fps
1 sec = 106 μ s

$$\text{Refresh rate} = 60 \text{ Hz} = \frac{1}{60} \text{ sec to scan} = 16.7 \text{ ms}$$

$$\text{The time for horizontal retrace} = 1024 \times 20 = 20480 \mu\text{s}$$

$$\text{The time for vertical retrace} = 20 \mu\text{sec}$$

$$\text{Total time spent for retrace} = 20480 + 20 = 20500 \mu\text{s} = 20.5 \text{ ms}$$

$$\therefore \text{The fraction of the total refresh time frame spent in retrace} = \frac{20.5}{16.7} = 1.227.$$

$$(x-2)^2 + (y-3)^2 = 25$$

$$\text{Comparing with } (x-h)^2 + (y-k)^2 = r^2$$

$$\text{we get, } (h,k) = (2,3)$$

$$\text{Radius} = 5$$

Case 1

$$d_i < 0$$

$$d_{i+1} = d_i + 2x_i + 1 + 1$$

$$x_{i+1} = x_i + 1$$

$$y_{i+1} = y_i$$

$$\text{Initial decision parameter} = 1 - r$$

$$= 1 - 5$$

$$= -4$$

Case 2

$$d_i \geq 0$$

$$d_{i+1} = d_i + 2(x_{i+1} - y_{i+1} + 1)$$

$$x_{i+1} = x_i + 1$$

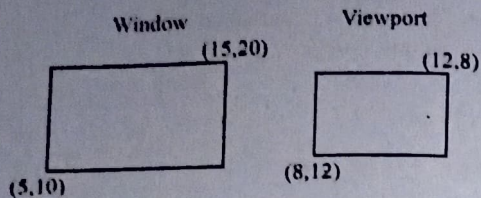
$$y_{i+1} = y_i - 1$$

$$\text{Starting point } (0,r) = (0,5)$$

n	d _k	x _{i+1}	y _{i+1}	Plot	(2,3) centre
0	-4	1	5	1,5	3,8
1	-4 + 2 × 1 + 1 = -1	2	5	2,5	4,8
2	-1 + 2 × 2 + 1 = 4	3	4	3,4	5,7
3	4 + 2 × 3 - 2 × 4 + 1 = 3	4	3	4,3	6,6

$x > y$ we can stop.

Remaining co-ordinate can find using symmetric property.



Here,

$$x_{wmin} = 5$$

$$x_{wmax} = 15$$

$$y_{wmin} = 10$$

$$y_{wmax} = 20$$

$$x_{vmin} = 8$$

$$x_{vmax} = 12$$

$$y_{vmin} = 12$$

$$y_{vmax} = 8$$

Step 2: Scaling of window to viewport is performed and for this we need scaling factor

$$S_x = \frac{\text{width of viewport}}{\text{width of window}}$$

$$= \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$$

$$= \frac{12 - 8}{15 - 5}$$

$$= \frac{4}{10} = \frac{2}{5}$$

$$S_y = \frac{\text{height of viewport}}{\text{height of window}}$$

$$= \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$$

$$= \frac{8 - 12}{20 - 10}$$

$$= \frac{-4}{10} = \frac{-2}{5}$$

$$\text{Scaling matrix} = \begin{bmatrix} \frac{2}{5} & 0 & 0 \\ 0 & -\frac{2}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 1: The object together with window is translated in such a way that left lower corner of window is shifted to origin.

Here, $T_x = 5$, $T_y = 10$

$$\text{Then translated matrix} = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 3: We translate the viewport to desired location on the screen.

$$\text{Then the translation matrix} = \begin{bmatrix} 1 & 0 & x_{vmin} \\ 0 & 1 & y_{vmin} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence, the viewing transformation matrix will be

$$T_{vv} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{2}{5} & 0 & 0 \\ 0 & -\frac{2}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculate yourself.

2018 spring

Calculate the access time for a pixel and a row for a graphics system having resolution of 1024*640 and frequency of 60 Hz.

Rasterize the points of given line end points A(-2,-4) and B(-6,-9) using Bresenham's line drawing algorithm.

Apply Cohen sutherland line clipping algorithm for calculating the saved portion of a line from (2,7) to (8,12) in a window ($x_{wmin} = y_{wmin} = 5$ and $x_{wmax} = y_{wmax} = 10$).

Calculate (x,y) coordinate of Bezier curve described by the following 4 control points (0,0), (1,2), (3,3), (4,0). Assume any needed values. Explain the Gouraud shading method with its advantages.

Solution:

Here, resolution = 1024*640

Frequency = 60 Hz Say 60 fps

That means system contains 640 scan lines and each line contains of 1024 pixels and refresh rate = 60 fps

1 frame takes $\frac{1}{60}$ sec.

1 from consists of 640 scan lines

640 scan lines takes $\frac{1}{60}$ sec.

∴ 1 scan lines take $\frac{1}{60 \times 640} = 0.00002604167 \text{ sec} = 26041.67 \text{ ns}$

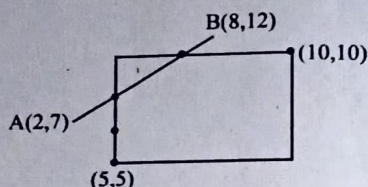
Since 1 frame = 1024×640 pixel = 655360 no. of pixel

So 1 frame needs $\frac{1}{60} \text{ sec.}$

655360 pixel needs $\frac{1}{60} \text{ sec.}$

1 pixel needs $\frac{1}{60 \times 655360} \text{ sec.} = 0.00000002543 \text{ sec.} = 25.43 \text{ ns}$

When,



A(2,7), B(8,12)

(x_1, y_1) (x_2, y_2)

$$\text{Slope (m)} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{12 - 7}{8 - 2} = \frac{5}{6}$$

$$x_{wmin} = y_{wmin} = 5$$

$$x_{wmax} = y_{wmax} = 10$$

Step 1: Region code for A and B

A(2,7)	B(8,12)
$2 < 5 = T 1 L$	$8 < 5 = F 0$
$2 > 10 = F 0 R$	$8 > 10 = F 0$
$7 < 5 = F 0 B$	$12 < 5 = F 0$
$7 > 10 = F 0 T$	$12 > 10 = T 1$

∴ Region code of A = 0001

Region code of B = 1000

Step 2: Logical anding of A and B

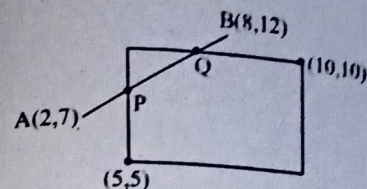
A = 0001

B = 1000

0000

This is the case of partially visible it might have intersected the window to some points.

Step 3:



We have to find the intersection point of line AB

For P:

Since P cuts the window cut left side so its x_{wmin} remains constant and y changes $x_{wmin} = 5$ to find

So, A(2,7) and P(5,y)

So, slope of AP = Slope of AB.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{5}{6}$$

$$\text{or, } \frac{y - 7}{5 - 2} = \frac{5}{6}$$

$$\text{or, } \frac{y - 7}{3} = \frac{5}{6}$$

$$\text{or, } 2y - 14 = 5$$

$$\text{or, } y = \frac{19}{2}$$

$$\therefore P(5, 9.5)$$

For Q:

Since Q cuts the window at top side so its y_{wmax} remains constant and x change

$$y_{wmax} = 10$$

x to find

A(2,7) and Q(x,10)

Slope of AQ = Slope of AB

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{5}{6}$$

$$\text{or, } \frac{10 - 7}{x - 2} = \frac{5}{6}$$

$$\text{or, } \frac{3}{x-2} = \frac{5}{6}$$

$$\text{or, } 18 = 5x - 10$$

$$\text{or, } \frac{28}{5} = x$$

$$\text{or, } x = 5.6$$

$$\therefore Q(5.6, 10)$$

Hence the line PQ is visible and the line AP and QB were clipped away or cutoff.

2018 spring (Belzer)

5. (b) Calculate (x, y) coordinate of Bezier curve described by the following 4 control points (0, 0), (1, 2), (3, 3), (4, 0). Assume any needed values.

Solution:

Control point = 4

Degree of polynomial = $n - 1$

We know the = 3

Bezier cubic polynomial equation,

$$P(t) = (1-t)^3 P_0 + 3t(1-t)^2 + 3t^2(1-t)P_2 + t^3 P_3 \quad 0 \leq t \leq 1$$

Control points:

$$A(0,0) = P_0(x_0, y_0)$$

$$B(1,2) = P_1(x_1, y_1)$$

$$C(3,3) = P_2(x_2, y_2)$$

$$D(4,0) = P_3(x_3, y_3)$$

Let us takes $t = 0, 0.25, 0.5, 0.75$

Using matrix method

$$P(t) = \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_0 \\ P_1 \\ P_2 \\ P_3 \end{bmatrix}$$

At $t = 0$

$$P(0) = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 0 \\ 3 & 3 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 & -3 & 0 \\ 3 & -3 & 0 \\ 3 & 6 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

At $t = 0.25$

$$P(0.25) = \begin{bmatrix} 0.25^3 & 0.25^2 & 0.25 & 1 \end{bmatrix} \begin{bmatrix} -2 & -3 & 0 \\ 3 & -3 & 0 \\ 3 & 6 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.90625 & 1.265625 & 0 \end{bmatrix}$$

At $t = 0.5$

$$P(0.5) = \begin{bmatrix} 0.5^3 & 0.5^2 & 0.5 & 1 \end{bmatrix} \begin{bmatrix} -2 & -3 & 0 \\ 3 & -3 & 0 \\ 3 & 6 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1.875 & 0 \end{bmatrix}$$

At $t = 0.75$

$$P(0.75) = \begin{bmatrix} 0.75^3 & 0.75^2 & 0.75 & 1 \end{bmatrix} \begin{bmatrix} -2 & -3 & 0 \\ 3 & -3 & 0 \\ 3 & 6 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3.09375 & 1.546875 & 0 \end{bmatrix}$$

t	x(t)	y(t)
0	0	0
0.25	0.90625	1.265625
0.5	2	1.875
0.75	3.09375	1.546875

2018 Fall

Define resolution and persistence. How long would it take to load a 640×480 frame buffer with 12 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load 24 bits per pixel frame buffer with a resolution of 1280×1024 with using the same transfer rate?

What will be the final coordinates of a triangle with vertices $A(2,3)$, $B(3,3)$, $C(3,2)$ after reflecting it about the line $y = x$?

Show how to use a 3 Dimensional matrix to rotate a unit cube about the axis defined by vector $(1,1,1)$.

Solution:

$Y=x$ 1st part is in slide

Slide 1

$$P' = C_m \times P$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 3 \\ 3 & 3 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 3 & 2 \\ 2 & 3 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore A'(3,2), B'(3,3), C'(2,3)$$

Here, for system 1.

$$\text{Resolution} = 640 \times 480$$

$$\text{bits} = 12 \text{ bits}$$

$$\text{Transferred rate} = 10^5$$

$$\begin{aligned} \text{Frame buffer size} &= 640 \times 480 \text{ pixels} \\ &= 640 \times 480 \times 12 \text{ bits} \\ &= 3686400 \text{ bits} \end{aligned}$$

$$\text{Since } 10^5 \text{ bits takes 1 sec.}$$

$$3686400 \text{ bits takes } \frac{3686400}{10^5} = 36.864 \text{ sec}$$

For system 2,

$$\text{Resolution} = 1280 \times 1024$$

$$\text{bits} = 24 \text{ bits}$$

$$\text{Transferred rate} = 10^5$$

$$\text{Frame buffer size} = 1280 \times 1024 \times 24 \text{ bits}$$

$$= 31457280 \text{ bits}$$

$$\text{Since } 10^5 \text{ bits takes 1 sec}$$

$$31457280 = \frac{31457280}{10^5} \text{ sec} = 314.5728 \text{ sec}$$

2017 Spring

2. (a) Define resolution. Suppose RGB raster system to be designed using on 8 inch $\times$ 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for frame buffer?

Solution:

$$\text{Resolution} = 8 \text{ inch} \times 10 \text{ inch [100 pixel per inch]}$$

First we convert it in pixel

$$\begin{aligned} \text{So resolution} &= 8 \times 100 \text{ by } 10 \times 100 \\ &= 800 \times 1000 \text{ pixel.} \end{aligned}$$

1 pixel can store 6 bits

$$\text{So, frame buffer size required} = 800 \times 1000 \times 6$$

$$= \frac{800 \times 1000 \times 6}{8} \text{ bytes}$$

$$= 6 \times 10^5 \text{ bytes}$$

2. (b) Digitize one octant of a circle by using midpoint circle generation algorithm center at origin and radius is 12.

Solution:

Digitize a circle

Here radius = 12

Origin = (0,0)

Using midpoint algorithm

We always start from (0,r)

$$\therefore x_0 = 0$$

$$y_0 = 12$$

$$\text{Initial decision parameter } (d_0) = 1 - r = 1 - 12 = -11$$

Case again

$$< x_{i+1} \ y_i$$

$$d_i + x_{i+1} + 1$$

$$> x_{i+1} \ y_i - 1$$

$$d_i + 2x_i - 2y_i + 1$$

n	d_n	x_{n+1}	y_{n+1}	Plot
0	-11	1	12	(1,12)
1	$-11 + 2 \times 1 + 1 = -8$	2	12	(2,12)
2	$-8 + 2 \times 2 + 1 = -3$	3	12	(3,12)
3	$-3 + 2 \times 3 + 1 = 4$	4	11	(4,11)
4	$4 + 2 \times 4 - 2 \times 11 + 1 = -9$	5	11	(5,11)
5	$-9 + 2 \times 5 + 1 = 2$	6	10	(6,10)
6	$2 + 2 \times 6 - 2 \times 10 + 1 = -1$	7	10	(7,10)
7	$-1 + 2 \times 7 + 1 = 14$	8	9	(8,9)
8	$14 + 2 \times 8 - 2 \times 9 + 1 = 13$	9	8	(9,8)

$x > y$ we can stop

4. (a) A mirror is placed vertically such that it passes through the points (10,0) and (0,10). Find the reflected view of triangle ABC with coordinates A(5,50), B(20,40) and C(10,70).

Solution:

A(5,50), B(20,40), C(10,70)

Here, $x_1 = 0$, $x_2 = 10$

$y_1 = 10$, $y_2 = 0$

$$\text{Slope (m)} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 10}{10 - 0} = -1$$

$$\therefore \text{Rotation angle } (\theta) = \tan^{-1}(m) = \tan^{-1}(-1) = -45^\circ$$

Steps:

1. Translation by $T_x = -10$

or, $T_x = 0$

$T_y = 0$

or, $T_y = -10$

both are possible

$$T_{(-10,0)} = \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Counter clockwise with $\theta = 45^\circ$

$$R_{\theta=45^\circ \text{CCW}} = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Reflection x-axis

$$\text{Ref}_{(x\text{-axis})} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Reverse rotation $R_{\theta=45^\circ}^{-1}$

$$R_{\theta=45^\circ \text{CW}}^{-1} = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Anti-translation by $T_x = 10$, $T_y = 0$

$$T_{(10,0)}^{-1} = \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{Cm} = T_{(10,0)}^{-1} R_{\theta=45^\circ \text{CW}}^{-1} \text{Ref}_{(x\text{-axis})} R_{\theta=45^\circ \text{CCW}} T_{(-10,0)}$$

$$= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

On solving

$$= \begin{bmatrix} 0 & -1 & 10 \\ -1 & 0 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, the required co-ordinate

$$P' = \text{Cm} \times P$$

$$= \begin{bmatrix} 0 & -1 & 10 \\ -1 & 0 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 20 & 10 \\ 50 & 40 & 70 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -40 & -30 & -60 \\ 5 & -10 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

2017 Fall

2. (b) Using the Bresenham's line drawing algorithm predict the pixels on the line from (2,2) to (12,10)

Solution:

End points A(2,2), B(12,10)

$$x_1 = 2, x_2 = 12$$

$$y_1 = 2, y_2 = 10$$

$$\text{Slope (m)} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10 - 2}{12 - 2} = \frac{8}{10} = 0.8 < 1$$

$$\Delta x > \Delta y$$

Case I

$$\Delta x = 10, 2\Delta x = 20$$

$$\Delta y = 8, 2\Delta y = 16$$

Initial decision parameter

$$P_0 = 2\Delta y - \Delta x = 16 - 10 = 6$$

n	P _k	X _{k+1}	Y _{k+1}
-	-	2	2
0	4	3	3
1	0	4	4
2	-4	5	4
3	12	6	5
4	8	7	6
5	4	8	7
6	0	9	8
7	-4	10	8
8	12	11	9
9	8	12	10

3. (b) Derive the composite transformation matrix for reflection of an object about a line $Y = mx + c$. Apply the derived matrix for the object A(4,2), B(7,3), C(9,2), D(10,1) on to the line $y = 3x$.

Solution:

$$y = 3x$$

Object A(4,2), B(7,3), C(9,2), D(10,1)

$$y = 3x$$

Here, $y = mx + c$

$$c = 0, m = 3$$

$$\tan \theta = 3, \theta = \tan^{-1}(3)$$

$$\tan \theta = \frac{P}{b}$$

$$\therefore P = 3, b = 1$$

$$h = \sqrt{P^2 + b^2} = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$\therefore \sin \theta = \frac{P}{h} = \frac{3}{\sqrt{10}}, \cos \theta = \frac{b}{h} = \frac{1}{\sqrt{10}}$$

$$C_m = R_0^{-1} \tan^{-1}(3) \text{CCW} R_{x\text{-axis}} R_{\theta \tan^{-1}(3) \text{CW}}$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} & 0 \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} & 0 \\ -\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

= Calculate yourself

$$P' = C_m \times P$$

$$= C_m \times \begin{bmatrix} 4 & 7 & 9 & 10 \\ 2 & 3 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

- Q. Prove that two successive rotations are additive to show that the composition of two successive rotations are additive i.e. $R(\theta_1).R(\theta_2) = R(\theta_1 + \theta_2)$.

Solution:

The rotation matrix R is given as,

$$R = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

we can write rotation matrix $R(\theta_1)$ as

$$R(\theta_1) = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 \\ -\sin\theta_1 & \cos\theta_1 \end{bmatrix} \text{ and } R(\theta_2) = \begin{bmatrix} \cos\theta_2 & \sin\theta_2 \\ -\sin\theta_2 & \cos\theta_2 \end{bmatrix}$$

$$R(\theta_1) \cdot R(\theta_2) = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 \\ -\sin\theta_1 & \cos\theta_1 \end{bmatrix} \times \begin{bmatrix} \cos\theta_2 & \sin\theta_2 \\ -\sin\theta_2 & \cos\theta_2 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_1 \cos\theta_2 + \sin\theta_1 (-\sin\theta_2) & \cos\theta_1 \sin\theta_2 + \sin\theta_1 \cos\theta_2 \\ -\sin\theta_1 \cos\theta_2 + \cos\theta_1 (-\sin\theta_2) & -\sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\theta_1 + \theta_2) & \sin(\theta_1 + \theta_2) \\ -\sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{bmatrix}$$

Since,

$$\cos(\theta_1 + \theta_2) = \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2$$

$$\sin(\theta_1 + \theta_2) = \cos\theta_1 \sin\theta_2 + \sin\theta_1 \cos\theta_2$$

2016 Spring

1. (b) Define resolution and image aspect ratio. A laser printer is capable of printing two pages (size 9×11 inch) per second at resolution of 600 pixels per inch. How many bits per second does such device required? (Assume 1 pixel = n bits)?

Solution:

Storage required per page in pixel

$$= 9 \times 11 \text{ inch/sec}$$

$$= 9 \times 11 \times 600 \times 600 \text{ pixel sec.}$$

$$\therefore 1 \text{ inch} = 600 \text{ pixels}$$

Since laser printer is capable of printing 2 pages per sec.

$$\therefore \text{bits per sec. required by}$$

$$\text{the printer} = 2 \times 9 \times 11 \times 600 \times 600$$

$$= 71280000$$

$$= 7.128 \times 10^7 \text{ pixel/sec.}$$

$$= 7.128 \times 10^7 \text{ n bits/sec. (Assume 1 pixel = n bits)}$$

3. (a) What will be the final coordinates of a polygon with vertices A(3,4), B(5,4), C(5,2), D(3,4) after it is reflected about a line $y = 2x + 1$?

Solution:

$$A(3,4), B(5,4), C(5,2), D(3,4)$$

$$\text{Line } y = 2x + 1$$

Hint,

$$m = 2$$

$$\theta = \tan^{-1}(2)$$

$$C_m = T_{(0,1)}^{-1} R_{\theta=63.43^\circ(CW)}^{-1} \text{Ref}_{x=200} R_{\theta=63.43^\circ(CW)} T_{(0,-1)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculate yourself

$$\therefore P' = C_m \times P = C_m \times \begin{bmatrix} 3 & 5 & 5 & 3 \\ 4 & 4 & 2 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2016 Fall

3. (b) A point (5,3) is required to be rotated by 45 degrees in clockwise direction and then scaled by a factor of 3, what will be the final transformed position after applying these transformations.

Solution:

$$\text{Point } P(5,3), Q = 45^\circ(CW), S_x = S_y = 3$$

About origin so,

$$C_m = S_{(3,3)} \cdot R_{\theta=45^\circ(CW)}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

= Calculate yourself

$$\therefore P' = C_m \times P = C_m \times \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}$$

1. (b) Consider two raster systems with resolution of 640 by 480 and 1280 by 1024. How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second? What is the access time per pixel in each system?

Solution:

Case 1:

Resolution = 640×480

No. of pixels in one frame = $640 \times 480 = 307200$

Since controller can access 60 frames in one second

Therefore, total no. of pixels accessed = 60×307200
 $= 18432000$ per sec.

Access time/pixel = $1/\text{total pixels accessed per sec.}$
 $= 5.4 \times 10^{-8}$ sec/pixel.

Case 2:

When resolution = 1280×1024

Total no. of pixels accessed by the raster system = $60 \times 1280 \times 1024$
 $= 78643200$ per sec

Access time per pixel = $1/78643200 = 1.2 \times 10^{-8}$ sec/pixel

2015 Spring

3. (a) Digitize a circle centered at (100,200) and having radius 8.

Solution:

Centre (100,200)

Radius = 8

Starting point (0,r) = (0,8)

Initial decision parameter = $1 - r = 1 - 8 = -7$

n	d_k	x_{i+1}	y_{i+1}	Plot	(100,200) centre
0	-7	1	8	(1,8)	101,208
1	$-7 + 2 \times 1 + 1 = -4$	2	8	2,8	102,208
2	$-4 + 2 \times 2 + 1 = 1$	3	7	3,7	103,207
3	$-1 + 2 \times 3 - 2 \times 7 + 1 = -6$	4	7	4,7	104,207
4	$-6 + 2 \times 4 + 1 = 3$	5	6	(5,6)	105,206
5	$3 + 2 \times 5 - 2 \times 6 + 1 = 2$	6	5	(6,5)	106,205

$x > 5$ stop

3. (b) What will be the final coordinates of a triangle with vertices A(2,3), B(3,3), C(3,2) after rotating it a by 45 degrees in anticlockwise direction then shifting it down by 3 units and finally enlarging it by twice its original size?

Solution:

A(2,3), B(3,3), C(3,2)

$\theta = 45^\circ$ (CCW)

$T_y = 3$ units

$S_x = S_y = 2$

$C_m = S_{(2,3)} T_{(0,-3)} R_{\theta=45^\circ}$ (CCW)

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculate yourself

$$\therefore P' = C_m \times P = C_m \times \begin{bmatrix} 2 & 3 & 3 \\ 3 & 3 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

2015 Fall

3. (a) Find the raster position along the region 1 of the ellipse path in first quadrant. The semi major and semi minor axes are 8 & 7 respectively and the center is (0,0).

See previous example

2014 Spring

4. (a) Perform a 45 degree rotation of a line A(8,3) and B(14,10):

i. About the origin.

ii. About a fixed point (4,2).

Solution:

A(8,3), B(14,10), $\theta = 45^\circ$ Let Say (CCW)

i. About origin

$$P' = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 8 & 14 \\ 3 & 10 \\ 1 & 1 \end{bmatrix}$$

ii. About fixed point (4,2)

$$C_m = T_{(-4,-2)}^{-1} R_{\theta=45^\circ} T_{(4,2)}$$

$$= \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculate yourself.

$$\therefore P' = C_m \times P = C_m = \begin{bmatrix} 8 & 14 \\ 3 & 10 \\ 1 & 1 \end{bmatrix}$$

4. (b) Reflect a rectangle A(2,2), B(5,2), C(2,4), D(5,4) about a line $x = y$.

Solution:

A(2,2), B(5,2), C(2,4), D(5,4)

Line $x = y$

We know when $x = y$

$$\text{Reflection matrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore P' = C_m \times P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 5 & 2 & 5 \\ 2 & 2 & 4 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} = \text{Calculate yourself}$$

2014 Fall

4. (a) Perform a 45 degree rotation of a line A(5,3) and B(10,15) about the origin.

Solution:

A(5,3), B(10,15)

$\theta = 45^\circ$

Let Say CCW

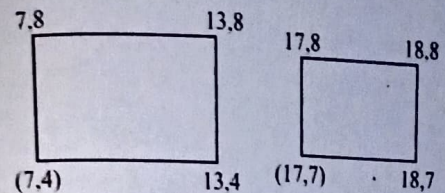
$$\begin{aligned} \therefore P' &= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 10 \\ 3 & 15 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 10 \\ 3 & 15 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 10 \\ 3 & 15 \\ 1 & 1 \end{bmatrix} \end{aligned}$$

= Calculate yourself

(A) OR

Calculate viewing transformation matrix with given information: given triangle with sides A(5,5), B(15,5), C(10,10), given window coordinate (7,4)(13,4)(13,8)(7,8) and view port location is (17,7)(18,7)(18,8)(17,8)?

Solution:



Step 1: The object together with window is trashed in such a way that left lower corner of window is shifted to origin.

$$T_x = -7, T_y = -4$$

$$\text{Then, translation matrix} = \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2: Scaling of window to viewport is performed and for this we need scaling factor.

$$S_x = \frac{\text{width of viewport}}{\text{width of window}} = \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}}$$

$$S_y = \frac{\text{height of viewport}}{\text{height of window}} = \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}}$$

Here,

$$X_{vmax} = 18$$

$$X_{vmin} = 17$$

$$X_{wmin} = 7$$

$$X_{wmax} = 13$$

$$S_x = \frac{18 - 17}{13 - 7} = \frac{1}{6}$$

$$Y_{vmin} = 7$$

$$Y_{vmax} = 8$$

$$Y_{wmin} = 4$$

$$Y_{wmax} = 8$$

$$S_y = \frac{8 - 7}{8 - 4} = \frac{1}{4}$$

$$\therefore \text{Scaling matrix} = \begin{bmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 3: We translate viewport to desired location on the screen.

$$\text{Then the translation matrix} = \begin{bmatrix} 1 & 0 & x_{vmin} \\ 0 & 1 & y_{vmin} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 17 \\ 0 & 1 & 7 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore T_{wv} = \begin{bmatrix} 1 & 0 & 17 \\ 0 & 1 & 7 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = Cm \times P = Cm \times \begin{bmatrix} 5 & 15 & 10 \\ 5 & 5 & 10 \\ 1 & 1 & 1 \end{bmatrix}$$

2013 Spring

4. (b) Scale a triangle A(0,0), B(1,1), C(3,2) by twice its original size, about origin and about point P(-1,-2).

Solution:

$$A(0,0), B(1,1), C(3,2)$$

$$S_x = S_y = 2$$

$$\text{Arbitrary point} = (-1, -2)$$

$$Cm = T_{(-1,-2)}^{-1} S_{(2,2)} T_{(3,2)}$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

= Calculate yourself.

$$\therefore P' = Cm \times P = Cm \times \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

2013 Fall

1. (a) Rotate the triangle A(2,3), B(5,3) and C(3,1) about a fixed point by 30°.

Solution:

$$A(2,3), B(5,3), C(3,1)$$

$$\theta = 30^\circ \text{ Let say CCW.}$$

Fixed point is not given so let say A be the fixed point.

$$Cm = T_{(2,3)}^{-1} R_{\theta=30^\circ \text{CCW}} T_{(-2,-3)}$$

$$= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ & 0 \\ \sin 30^\circ & \cos 30^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

= Calculate yourself.

$$\therefore P' = Cm \times P = Cm \times \begin{bmatrix} 2 & 5 & 3 \\ 3 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$